

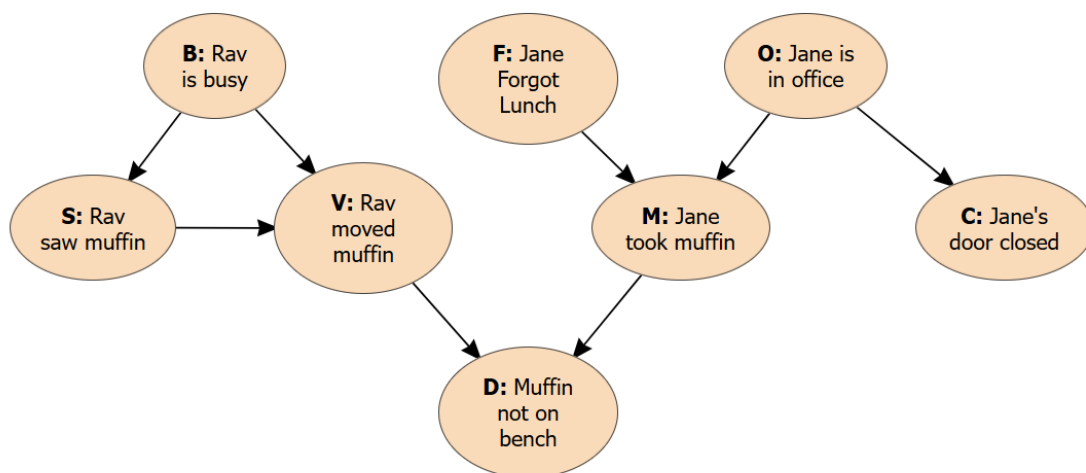
Monash University, Faculty of Information Technology

FIT3080

Lab: Bayesian Networks – d-separation, Inference & Decision Networks

Exercise 1: d-separation (& d-connection)

Scenario: You've brought a muffin to work and accidentally left it out on the bench before heading out for an appointment. Jane has a habit of taking food that's out if she's in the office and has forgotten her lunch. Rav will move the muffin somewhere to keep it safe for you, but only if he sees it and he's not busy. While waiting at your appointment, you sketch the following model:

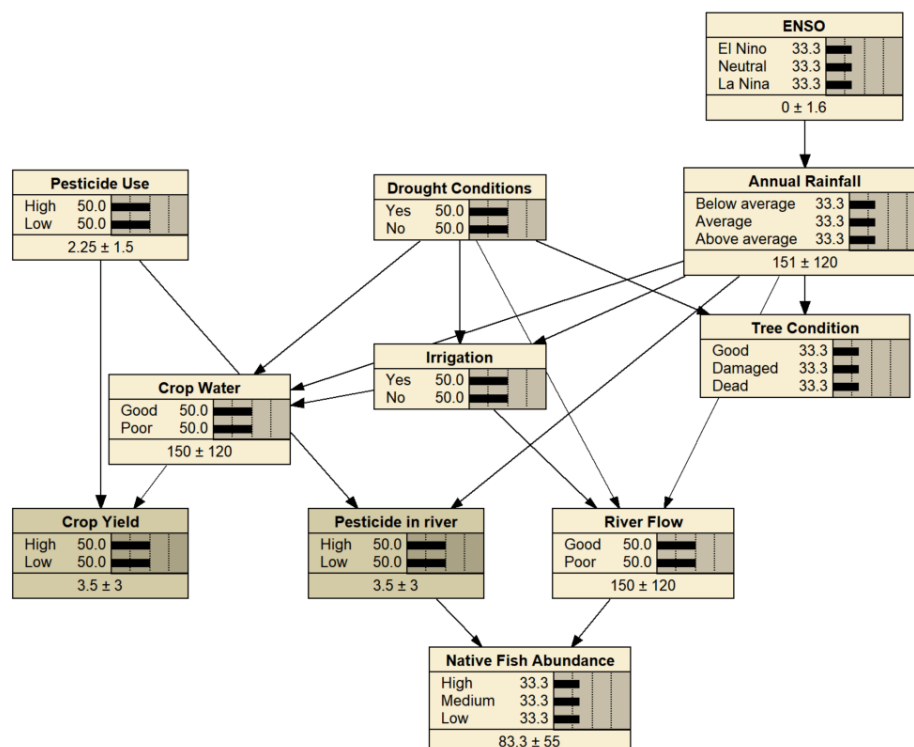


- What nodes are d-separated from D?
None (they are all d-connected to D)
- What nodes are d-separated from D given you observe M?
F, O and C
- Is B d-separated or d-connected to D when you observe S and M? Why?
It is d-connected. This is because there is an unblocked path between B and D, $B \rightarrow V \rightarrow D$ (a chain with no evidence on it).
- What nodes are d-connected to C?
O, M and D
- What nodes would be d-connected to C if you observed M?
O and F
- Are F and O d-connected if you observe i) D, ii) V, iii) D and V?
i) Yes
ii) No
iii) Yes
- (Optional) Discuss the answers in (f), in the context of the problem. For example, why does observing whether or not there's a muffin on the bench (D) change the probabilistic relationship between Jane forgetting her lunch (F) and Jane being in the office (O)?
Observing whether there's a muffin on the bench (D) provides information about

whether Jane took the muffin (M). If you know whether or not Jane took the muffin, all the reasons (causes) for that observation become related. Specifically, knowing something about one of M's causes, e.g., Jane being in the office, will either explain away or support the other of M's causes (Jane forgetting lunch).

Exercise 2: Learning, reasoning and evaluation

An extended version of the native fish model is shown below:

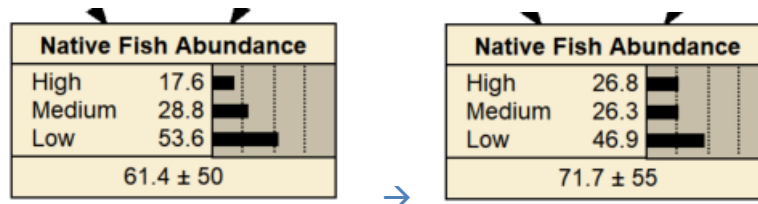


This model contains some nodes where numeric intervals have been provided for the states of the node. In addition to the usual probabilities, Netica will display an expected value for these nodes (e.g. 3.5 ± 3 for Crop Yield) at the bottom of the node.

You've been given three datasets that have been prepared to work with this BN. Both the unparameterised model (**nfv3_unparameterised.dne**) and the datasets (**nfv3_complete.csv**, **nfv3_missing.csv** and **nfv3_testing.csv**) can be downloaded via Moodle, in the applied class section for Week 7. (Look for "Week 7 Applied Class: Materials for Q2 - BNs and datasets".)

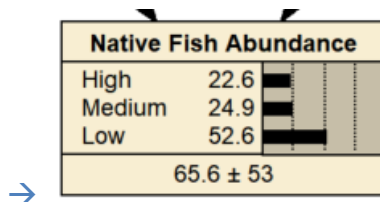
- a) Train the given model on **nfv3_complete.csv** using count learning, by using Netica's count learning function (menu: Cases -> Learn -> Incorp Case File. Leave the 'Enter degree' prompt at 1)
 - i. How does native fish abundance – both the *probability distribution and the expected value* – change when you observe low pesticide use?
 The probability distribution changes from 0.176,0.288,0.536 to 0.268,0.263,0.469 (e.g. the probability of 'high' fish abundance increases

from 18% to 27%), and the expected value increases from 61.4 +/- 50 to 71.7 +/- 55.



- ii. How does it change when you also know that tree condition is not good?
(Note: shift-click to set negative evidence in Netica)

The probability distribution shifts to 0.226, 0.249, 0.526 (i.e. the probability of high fish abundance goes down), and the expected value decreases to 65.6 +/- 53.



- b) Clear all evidence (menu: Cases -> Remove findings), and run a sensitivity to findings (menu: Network -> Sensitivity to findings) on the native fish abundance node.

- i. What 3 nodes have the strongest influence on native fish abundance? (Not including native fish abundance itself)

River flow, Irrigation and Annual Rainfall:

Sensitivity of 'Native Fish Abundance' to a finding at another node:					
Node	Variance Reduction	Percent	Mutual Info	Percent	Variance of Beliefs
Native Fish Abundance	2462	100	1.43996	100	0.3806454
River Flow	854.1	34.7	0.36242	25.2	0.0641095
Irrigation	93.41	3.79	0.03707	2.57	0.0081763
Annual Rainfall	68.67	2.79	0.04049	2.81	0.0085209
ENSO	50.68	2.06	0.02879	2	0.0062051
Drought Conditions	20.31	0.825	0.00789	0.548	0.0017766
Pesticide in river	14.98	0.608	0.01517	1.05	0.0010265
Tree Condition	11.26	0.458	0.00460	0.319	0.0010376
Pesticide Use	10.85	0.441	0.00395	0.274	0.0004641
Crop Water	0.1991	0.00809	0.00077	0.0536	0.0001465
Crop Yield	0.08171	0.00332	0.00073	0.0504	0.0000797

- ii. How does this change if river flow is good? What might explain the change?

Pesticide in River becomes the strongest influence (up significantly from 3rd

weakest influence). Annual Rainfall and ENSO are next highest, but Irrigation now becomes much less significant.

Irrigation becomes less significant because its primary influence was via River Flow (in a chain), which is now observed (blocking the chain). Observing River Flow (as an observed common effect) opens other paths, but they are weaker.

Pesticide in River was the weaker of two direct influences on Native Fish Abundance. It is now the only direct influence, and hence everything must go through Pesticide in River in order to influence Native Fish Abundance.

- c) Clear all evidence, and evaluate the performance of the model in predicting native fish abundance (menu: Cases -> Test With Cases) using **nfv3_complete.csv**.
- i. What is the error rate, confusion matrix and log score?

Error rate: 27.9%, Confusion matrix:

Confusion:

.....Predicted.....			
High	Medium	Low	Actual
-----	-----	-----	-----
98	64	9	High
46	161	81	Medium
9	70	462	Low

Log score: 0.6823

- ii. Why might using **nfv3_complete.csv** not be the best way to test the model?

1) It's the same data that we used to train the model and 2) it's complete (no missing values), so we *never* test anything other than the model's CPT for $P(\text{Native Fish Abundance} | \text{Pesticide in River and River Flow})$. Everything else is d-separated.

- iii. Now test the performance with **nfv3_testing.csv**. (50% of the values in this file are missing. You can inspect the file in a spreadsheet tool to see them.) What is the new error rate, confusion matrix and log score? How does it compare to the test that used **nfv3_complete.csv**?

Error rate: 36.09%, Confusion matrix:

Confusion:

.....Predicted.....			
High	Medium	Low	Actual
-----	-----	-----	-----
11	20	13	High
6	34	40	Medium
0	14	120	Low

Log score: 0.802

The metrics are worse. The error rate is higher and the log score is higher. The numbers in the confusion matrix are all lower because of the missing values. But the proportion of errors (the error rate) is of course higher. Particularly, when the prediction is 'Low', there is a higher error rate $(13+40)/(13+40+120) = 30.6\%$ compared to $(9+81)/(9+81+462) = 16.3\%$ previously.

- d) Remove all the CPTs (menu: Table -> Remove) and use count learning again to train the model with **nfv3_missing.csv**. (A high 80% of the values in this dataset are missing.) Test the newly trained model's ability to predict native fish abundance with **nfv3_testing.csv**.
- What is the error rate, confusion matrix and log score?

Error rate: 41.86%, Confusion matrix:

Confusion:

.....Predicted.....			
High	Medium	Low	Actual
-----	-----	-----	-----
23	0	21	High
21	0	59	Medium
7	0	127	Low

Log score: 0.9668

- What do the results in the confusion matrix suggest?

The model never predicts 'Medium' native fish abundance, even though 'Medium' actually occurs 80 times in the data $(21 + 59)$, making up $80/258 = 31\%$ of all cases in the data.

- How does this compare to the results from (c) (iii)? How would you explain the difference in the results?

The error rate is worse and log score are worse, but they don't look too much worse. The confusion matrix, however, suggests the model is behaving very badly for Medium native fish abundance cases. The trained probability

distribution for fish abundance only assigns ‘Medium’ a 16% chance, which may be due to too much missingness in the data to produce a useful result, at least with count learning.

- e) Repeat (d), but this time use expectation-maximisation (menu: Cases -> Learn -> Learn using EM; this may take 1-2 minutes). How do the results compare?

Error rate: 36.05%, Confusion matrix:

Confusion:

.....Predicted.....			
High	Medium	Low	Actual
-----	-----	-----	-----
20	16	8	High
19	24	37	Medium
3	10	121	Low

Log score: 0.8486

The performance is significantly better, and the model actually predicts ‘medium’ native fish abundance, with a similar error rate to (c)(iii). The overall rate is also similar, but the log score is slightly worse. EM performs much better with missing data than count learning.

(Note that the Netica demos in the recorded lecture may help you with Q2.)

Exercise 3: Decision Networks, Netica¹

Clare’s football team, Melbourne, is going to play her friend John’s team, Carlton. If the weather is wet, Melbourne has a 60% chance of winning, but if the weather is dry, Melbourne’s chances of winning are only 25%. At this time of the year, there is a 30% chance of rain on any day.

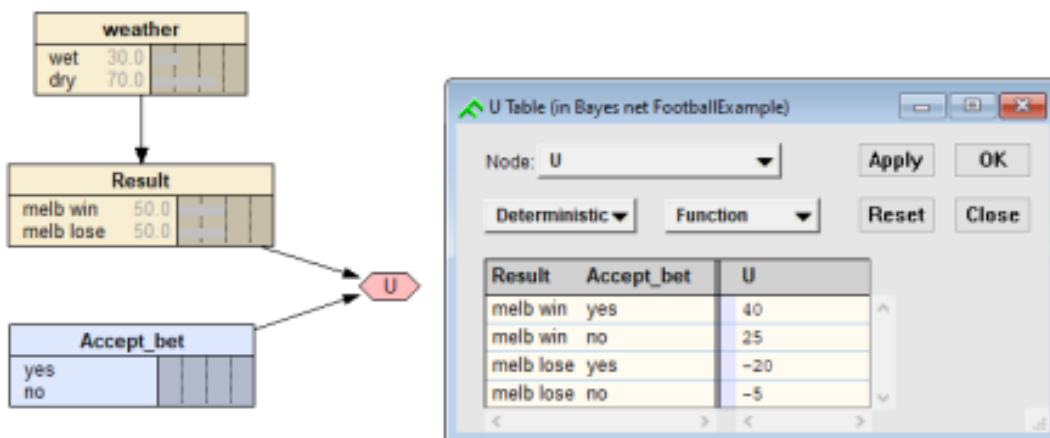
John offers Clare a friendly bet: whoever's team loses will buy the wine next time they go out for dinner (at a cost of \$15).

To decide whether to accept the bet, Clare will assess her team’s chances of winning. She also knows that she will be happy if her team wins and unhappy if her team loses, regardless of the bet.

¹From Korb & Nicholson, 2010 2nd Edition.

- Using the variables *Weather* and *Result* (for Melbourne) and the decision node *Accept* bet, **build a BDN for this scenario using Netica software**. Add utilities according to the narrative.
- Determine whether Clare should make the bet for each weather condition.

The utilities incorporate the value of Clare's happiness at winning (e.g. perhaps \$25, if utility is equated to dollars) and losing (\$-5), plus the gain from the bottle of wine if she accepts the bet and wins, and the loss of buying the bottle of wine (\$-15) if she accepts the bet and loses.



Exercise 4: Decision Networks, Netica²

Robert is trying to decide whether to study hard for the AI exam. He wants to get a High Distinction (HD) for the unit (he is not interested in any other marks), but he knows that his mark will depend not only on how hard he studies but also on how hard the exam is, and if he did well in the Theory of Computation unit (which indicates how well prepared he is for the current unit). There is a 50-50 chance that the AI exam will be hard, and 10% of all students receive an HD in Theory of Computation.

In addition, Robert would prefer to get a HD without studying hard.

- State which nodes are required in a Bayesian Decision Network, what are their types, and what values they should take.

Node name	Type	Values
TheoryComp	Chance (categories)	{HD,Other}
ExamHard	Chance (boolean)	{T,F}
AIResult	Chance (binary)	{HD,Other}
StudyHard	Decision	{Yes,No}

² Based on Korb & Nicholson, 2010 2nd Edition, 4.10, Exercise 3.

U	Utility	Numerical
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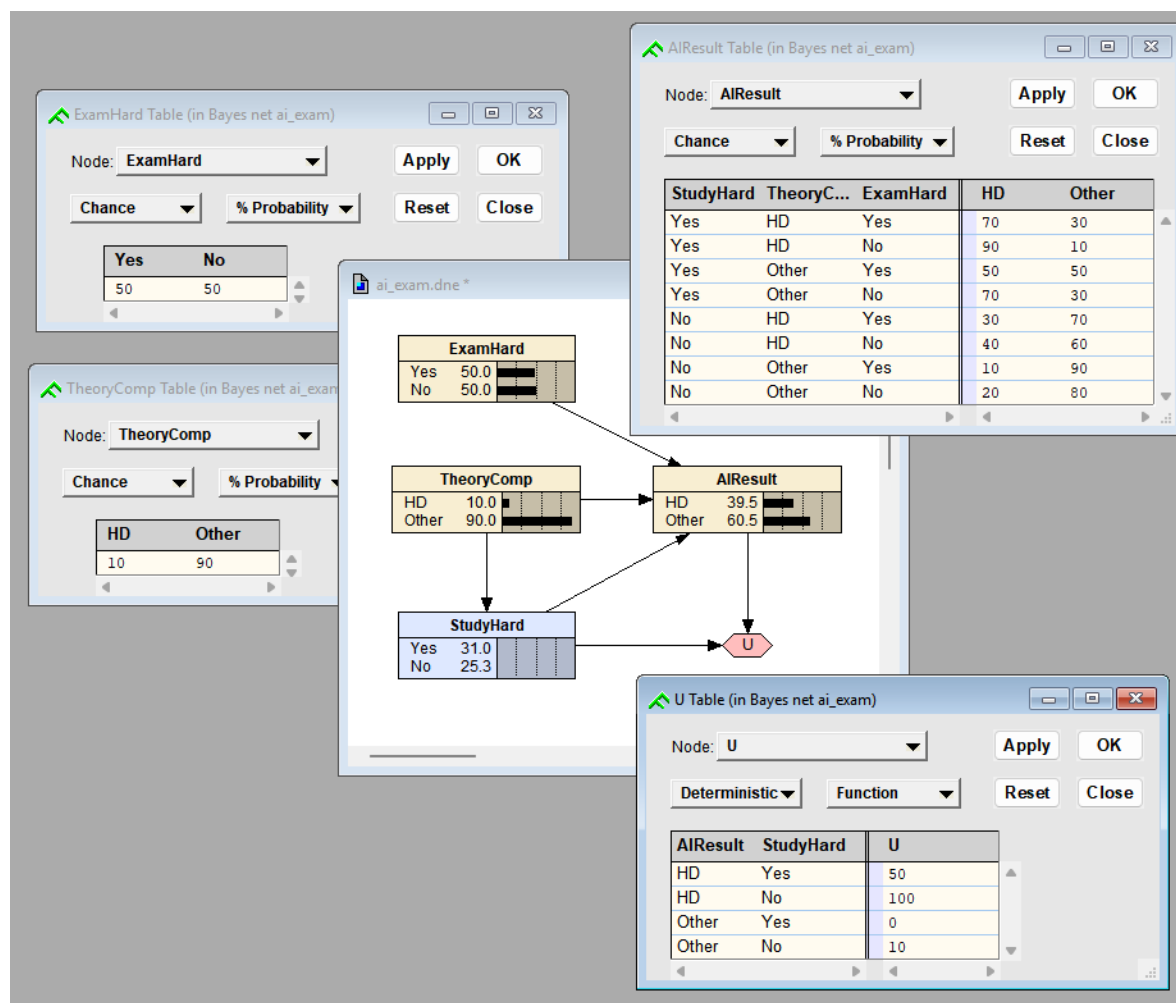
2. What is Robert's utility function? What chance nodes, if any, will it depend on? Does it depend on the decision node?

Robert's satisfaction will depend on both the subject result and how hard he studied, so $U = f(\text{StudyHard}, \text{AIResult})$. The best case is not studying hard and a good result, the worst case is studying hard and a bad result. We can arbitrarily assign a top and bottom utility of 100 and 0. Other cases would be in between, with a HD much more important than not studying hard. His utility function can be represented as a table, such as:

AIResult	StudyHard	Value
HD	Yes	50
HD	No	100
Other	Yes	0
Other	No	10

3. **Build a Decision network to model this problem using the Netica Software:** insert arcs that connect the chance, decision and utility nodes. Quantify the relationships in the network by populating the CPTs of chance nodes and the utility table for the utility node (use your own intuitions to come up with suitable values). Do you need an information link between the evidence node and the decision node?

A BN for this problem is shown below:



Since Robert will know his result in the Theory of Computation subject, *TheoryComp* will be an evidence node. If we already knew Robert's mark for Theory of Computation when building the BN, we could assume (or eliminate) this variable from the decision network, i.e., delete it and use only the CPT values for his actual mark, e.g., $\text{TheoryComp} = \text{Other}$. However, we will keep this node, its prior and the full CPT for *AIResult* to show the full story.

We need an information link to show the evidence should be available before the decision. The priors for *TheoryComp* shown above is the distribution over {HD, Other} for all students taking that subject. An alternative source for the prior would be to have it reflect Robert's marks over his university career to date, say $\text{Pr}(\text{TheoryComp}) = (0.4, 0.6)$. The former choice models the difficulty of the subject, the latter choice focuses on Robert's capabilities as a student.

4. Once you have built your model, use Netica to show the beliefs for the chance nodes and the expected decisions for the utilities before any evidence is added. Add the evidence regarding Robert's mark in Theory of Computation, and use Netica to see how the beliefs and the decision change, if at all. Explain your results.

Expected utilities for Robert's decision alternatives, and the associated decision table, without evidence and then given different marks for *TheoryComp* are shown in the table below (highest expected utility in each case in bold). Intuitively, Robert doesn't need to study hard if he received a HD in Theory of Computation, as he'll probably get a HD in AI anyway (although the expected utilities are very close). Otherwise, it is worthwhile for him to study hard.

TheoryComp	Pr(AIResult=HD)	EU(SH=Yes)	EU(SH=No)	Decision
No evidence	0.395	31.0	25.3	Yes
HD	0.575	40.0	41.5	No (just)
Other	0.375	30.0	23.5	Yes